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PICKLEBALL PADDLE INLAY WITH METALLIC LATTICE COMPOSITE STRUCTURE

Abstract

A paddle includes a core and at least one metallic lattice structure coupled to the core.

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Background/Summary

CROSS REFERENCE TO RELATED PATENT APPLICATIONS [0001] This application is a non-provisional application which claims priority from U.S. Provisional Patent Application No. 63/557,262, filed on Feb. 23, 2024, and incorporates the content thereof in its entirety by reference.

TECHNICAL FIELD

[0002] The present disclosure relates generally to sport equipment. Specifically, the present disclosure relates to paddles such as pickleball paddles that include metal lattice structures.

BACKGROUND

[0003] Pickleball is an indoor or outdoor paddle sport in which two (singles) or four (doubles) players hit a perforated, hollow plastic ball with paddles over a 34-inch-high net until one side is unable to return the ball or commits a fault by violating a rule.

[0004] As the sport of pickleball has advanced, the demand for improved and sophisticated equipment requires innovations that can handle competitive play. Some pickleball paddles on the market are not capable of providing play-enhancing features that increase the effectiveness of a player. Additionally, some of the equipment available is less durable than others, and under competitive use, it is important that the players have reliable and durable equipment to provide them with competitive use. Therefore, it is desirable to address a specific deficiency in the realm of pickleball sporting equipment and develop a high-performance pickleball paddle capable of enduring the rigorous demands of competitive play.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] The detailed description is set forth below with reference to the accompanying figures. In the figures, the left-most digit(s) of a reference number identifies the figure in which the reference number first appears. The use of the same reference numbers in different figures indicates similar or identical items. The systems depicted in the accompanying figures are not to scale and components within the figures may be depicted not to scale with each other.

[0006] FIG. 1 illustrates an exploded, isometric view of a paddle, according to an example of the principles described herein.

[0007] FIG. 2 illustrates a metallic lattice structure, according to an example of the principles described herein.

[0008] FIG. 3 illustrates a metallic lattice structure, according to an example of the principles described herein.

[0009] FIG. 4 illustrates a metallic lattice structure, according to an example of the principles described herein.

[0010] FIG. 5 illustrates a metallic lattice structure, according to an example of the principles described herein.

[0011] FIG. 6 illustrates a metallic lattice structure, according to an example of the principles described herein.

[0012] FIG. 7 illustrates a metallic lattice structure, according to an example of the principles described herein.

[0013] FIG. 8 illustrates a flow diagram of an example method of manufacturing a paddle, according to an example of the principles described herein.

DESCRIPTION OF EXAMPLE EMBODIMENTS

Overview

[0014] The present disclosure provides for a paddle with superior play performance. This may include a paddle design that significantly enhances player control, increases power output, and extends durability, without compromising other critical performance metrics. The combination of

advanced materials and composite structures has led to significant innovations in sports equipment including in pickleball paddles. In an embodiment, the paddle may include a core and a layer including a metallic lattice structure, which layer with the metallic lattice structure may be directly coupled to the core or indirectly coupled to the core. Furthermore, the metallic lattice structure may include a metal or metal alloy. For example, in an embodiment, the metallic lattice structure may include a titanium, a titanium alloy, or a treated metal.

[0015] In an embodiment, the paddle may further include an intermediate layer coupled to the core and the layer of the metallic lattice structure. In an embodiment, the paddle may further include at least one outer layer coupled to the at least one intermediate layer or the layer of the metallic lattice structure. Additionally, and/or alternatively, in an embodiment, the paddle may include at least one outer layer coupled to the at least one metallic lattice structure or the intermediate layer, if an intermediate layer is included.

[0016] In an embodiment, the layer with the metallic lattice structure may include at least one of a wireframe lattice structure, a dimpled lattice structure, a perforated lattice structure, an expanded lattice structure, a flattened expanded lattice structure, a corrugated lattice structure, a wire netting lattice structure, a wire-mesh lattice structure, or a woven mesh lattice structure. In an embodiment, the core may include a honeycomb architecture.

[0017] This disclosure describes a paddle and techniques for forming or manufacturing a paddle that includes a metallic lattice structure. The example paddles described herein introduce a reinforced metal lattice structure within a composite pickleball paddle. The inclusion of the metal lattice structure may enhance the durability, power, and control of play of the paddle. By leveraging the advanced materials and manufacturing techniques described herein, the paddle may offer a groundbreaking solution to improve player performance and satisfaction. Further, the paddle including the metal lattice structure may represent a major advancement in sports equipment design and sets a new standard for pickleball paddle construction. Embedding a reinforced metal lattice within composite pickleball paddles is a significant innovation in pickleball equipment. It utilizes advanced materials and composite construction techniques to produce paddles that may improve player performance and enjoyment.

[0018] Certain implementations and embodiments of the disclosure will now be described more fully below with reference to the accompanying figures, in which various aspects are shown. However, the various aspects may be implemented in many different forms and should not be construed as limited to the implementations set forth herein. The disclosure encompasses variations of the embodiments, as described herein. Like numbers refer to like elements throughout.

[0019] FIG. 1 illustrates an exploded, isometric view of a paddle **100** (e.g., a completed paddle ready for play, a paddle blank, an incomplete paddle for further processing), according to an example of the principles described herein. In other words, the use of the term paddle hereinafter, and as claimed, may refer to either a paddle that is considered completely finished and is ready for play (competitive or casual); or an incomplete paddle that is not in a finished state for play, but rather is only a stage of processing and has had some elements of a paddle, and therefore may be embodied as claimed hereinafter. Such an incomplete paddle may also be referred to as a paddle blank, which is ready for further finishing stages, perhaps in a different facility.

[0020] In an embodiment, the paddle **100** may include a core **102** as a base layer that serves as a foundational substrate portion of the paddle **100**. As discussed further hereinafter, the paddle **100** may include a first layer **104**, a second layer **106**, and/or a third layer **108**. In an embodiment, the paddle **100** may be formed in the specifically shown arrangement, with the first layer **104** placed on a surface **102a** of core **102**, which is sandwiched between the core **102** and the second layer **106**. In an embodiment, the second layer **104** may be sandwiched between the first layer **104** and the third layer **108**. However, in an embodiment (not shown explicitly), the first layer **104** may be placed on surface **102a** and/or a surface **102b** that is opposite surface **102a** on core **102**. Likewise with the placement of one or more second layers **106** and one or more third layers **108**. To be clear, in an

embodiment, the first layer **104**, the second layer **106**, and/or the third layer **108**, may be duplicated and placed on the opposite side of core **102**.

[0021] Furthermore, the body of the paddle **100** may include, for example, a handle portion **110** (“handle”), a neck portion **112** (“neck”), and a head portion **114** (“head”). The paddle **100** may be further defined by additionally narrowly defined sections or features known in existing paddles, including but not limited to, an outer perimeter that surrounds at least the central region of the playing face in the head portion, and the throat (at least a part of which may be part of the playable face and may extend part way into the handle region). In addition, the paddle **100** may further include one or more of an edge guard coupled to an edge of the paddle, pallets that form part of the handle, a grip, an end cap coupled to an end of the handle, and a rubber or silicone band coupled to an end of the handle opposite the end cap.

[0022] As discussed above, the metallic lattice structure may be applied to all or a portion of the paddle **100**. In an embodiment, although the first layer **104**, the second layer **106** and the third layer **108** are depicted in FIG. **1** as covering substantially all of the portion of the core **102**, in one example, the first layer **104**, the second layer **106** and/or the third layer **108** may cover less than an entirety of the surface area of the paddle **100**. For example, the first layer **104**, the second layer **106** and/or the third layer **108** may cover less than all of the handle **110**, the neck **112**, and the head **114** such as the just the head **114** and not the neck **112** of the handle **110**.

[0023] In an embodiment, the paddle **100** may include an intermediate layer coupled to the core **102** and the metallic lattice structure. In this example, the intermediate layer may be the first layer **104**. Further, in an embodiment, the metallic lattice structure may be embodied as the second layer **106** and/or the third layer **108**.

[0024] In an embodiment, the paddle **100** may include an intermediate layer coupled to the metallic lattice structure and an outer layer. In an embodiment, the intermediate layer may be embodied as the second layer **106**, the metallic lattice structure may be the first layer **104**, and the outer layer may be embodied as the third layer **108**.

[0025] In an embodiment, both the first layer **104** and the second layer **106** may include a metallic lattice structure. In an embodiment, a first metallic lattice structure may be embodied as the first layer **104** and may be coupled to the core **102**, and a second metallic lattice structure may be embodied as the second layer **106** and may be coupled to the first metallic lattice structure (e.g., the first layer **104**). The outer layer may be the third layer **108** and may be coupled to the second metallic lattice structure (e.g., the second layer **106**).

[0026] In an embodiment, the metallic lattice structure may be embodied as the outer layer such as the third layer **108**. However, as stated above, in an embodiment, the metallic lattice structure may be included as any of the first layer **104**, the second layer **106**, and/or the third layer **108**. For any of the first layer **104**, the second layer **106**, and the third layer **108** that does not include a metallic lattice structure, those layer(s) may include, for example, a woven or non-woven carbon fiber, a woven or non-woven fabric, a fiberglass, a polymer, a metal or metal alloy, a cork, a wood, or other materials. Further, in an embodiment, the outer layer such as the third layer **108** may include a fiberglass, a plastic, a metal or metal alloy, or other material.

[0027] The metallic lattice structure that may be included as the first layer **104**, the second layer **106**, and/or the third layer **108** and may form part of a composite layer of the paddle **100**. The composite layer of the paddle **100** may include, for example, the first layer **104**, the second layer **106** and/or the third layer **108**. Further, the composite layer of the paddle **100** may be formed as a whole before application to the paddle **100** including one or more of the first layer **104**, the second layer **106**, and/or the third layer **108**. The aspects of the above description applies to the composite layer of the paddle **100**.

[0028] In an embodiment, the core **102** may be a rigid material or a flexible material. In either case, the material of the core may be reinforced to provide a measure of, or an increase in, the stiffness/rigidity characteristic of the paddle **100** by adding one or more additional layers of

material. The “additional layers” may include any of those layers (**104, 106, 108**) described further hereinafter, or such stiffness may be provided by means other than via the additional layers (**104, 106, 108**) discussed in this disclosure. For example, though depicted as a solid, continuously consistent material in FIG. 1, the core **102** may include additional elements of structural formation for rigidity or strength, such as a composite of one or more materials and/or one or more architectural frameworks (e.g., shape structure, substructure, substructural frame, frame, skeleton, etc.). For example, in an embodiment, the core **102** may include a honeycomb architectural framework. A substructural frame may be found in a portion (e.g., a center region of the head, the entire head, the handle, the neck, the throat, a peripheral region of any of the aforementioned areas) of the core **102** of the paddle **100**, or throughout the core **102** (i.e., side to side, border to border, end to end, etc.). However, in light of the endless possibilities of the base architectures available, FIG. 1 depicts the core **102** as having a continuous playing surface with no visible underlying substructure. Further, the core **102** may be made of any suitable material or a composition of materials. In an embodiment, the material of the core **102** (either the entirety, a portion of, the architectural framework, and/or the filler material) may include, for example, one or more metals such as aluminum, metal alloys, polymers with or without composite fibers/fibrous material, such as Nomex®, etc.

[0029] As indicated above, the paddle **100** may include one or more of a plurality of layers of materials on at least one face of the paddle **100**. In an embodiment, the plurality of layers (**104, 106, 108**) may be applied to both a first face (e.g., surface **102a**) and a second face (e.g., surface **102b**) of the paddle **100** to provide the qualities of the plurality of layers described herein on both faces. In an embodiment, the first face of the paddle **100** may include fewer, equal, or more layers of material and/or the layers may be of different characteristics or materials as compared to the layer or layers (if any) on the second side. In other words, while the layers described herein are described as shown with respect to the first side of the paddle **100**, the layers may be similarly applied to the second side of the paddle as well.

[0030] In an embodiment, the first layer **104**, the second layer **106**, and/or the third layer **108** may include a substrate (e.g., sheet) of a metallic lattice structure, or of a non-metal (either in a lattice structure or as a continuous solid material) directly or indirectly coupled to the core **102**. For the purposes of this application, the term “lattice” may define any two-dimensionally extending structure formed using, or being shaped to appear as, strips or interconnected rows of material that are crossed, overlapping, woven, wrapped, affixed, aligned, dimpled, bubbled, entangled, etc. or otherwise engaged in any of a variety of manners, where the “strips” of material fastened (e.g., molded, melted, stuck, embedded, cast, fixed, arranged, stuck, pressed, etc.) together in a structure that has a sheet shape (e.g., extending two-dimensionally, with a predetermined thickness in the third dimension). That is, a layer (**104, 106, 108**) with a lattice may be formed with a three-dimensional direction thickness that is initially thicker than desired for the layer (**104, 106, 108**) to be implemented in the paddle **100**, but which may be reduced to a desired thickness. Alternatively, the lattice structure may be formed directly as the desired thickness. Although three layers are depicted in FIG. 1, including the first layer **104**, the second layer **106**, and/or the third layer **108**, more or fewer layers may be included in the paddle **100**.

[0031] In an embodiment, the first layer **104**, the second layer **106**, and/or the third layer **108** may include a metallic lattice structure as described herein. Thus, a layer with the metallic lattice structure may be directly coupled to the core **102**, as in the case of the first layer **104**, or indirectly coupled to the core **102**, as in the case of the second layer **106**, and/or as in the third layer **108**, which may be coupled directly to the second layer **106**.

[0032] In an embodiment, the metallic lattice structure may include a metal or metal alloy. For example, in an embodiment, the metallic lattice structure may be formed of a titanium or a titanium alloy, which may provide corrosion resistance, a superior strength-to-weight ratio, low density, high heat-resistance, high heat transfer capability, good oxidation capabilities, a high melting point, a

low elasticity module, excellent erosion resistance, low thermal expansion coefficient, etc., all of which assist in manufacturing a paddle that assist in providing superior quality during play. Although titanium and titanium alloys are described herein in connection with the metallic lattice structure, any material may be used in connection with the paddle **100**. In an embodiment, the metallic lattice structure may include a treated metal.

[0033] The metallic lattice structure may include any lattice structure or architecture that may assist in imparting structural and/or performance benefits to the paddle **100**. For example, in embodiment as seen in FIG. 2, the metallic lattice structure **200** may include a wireframe lattice structure including overlapping vertical wires **202** and horizontal wires **204** that make up the lattice structure **200**. For example, in an embodiment, the metallic lattice structure **200** of FIG. 2 may include a heavy wire mesh in which heavy gauged wires are formed and coupled at approximately 90-degree angles. The wire in the example metallic lattice structure **200** of FIG. 2 may have any gauge and may be spaced at any distance from one another.

[0034] In an alternative embodiment as seen in FIG. 3, the metallic lattice structure **300** may include a dimpled lattice structure **300** in which dimpling or other patterned or regular indentations **302** are formed in a surface of, for example, a metal sheet **304** (portions between the dimpling) to form a lattice structure **300** with regular dimpling along the surface of the metal sheet. The dimples **302** formed in the metallic lattice structure **300** may have any size, dimensions, and shapes including sizes and dimensions associated with an outer perimeter, depth, and other features of the dimples **302**.

[0035] In an alternative embodiment as seen in FIG. 4, the metallic lattice structure **400** may include a perforated lattice structure **400** in which a pattern of apertures **402** is formed in the surface of the metallic lattice structure **400** along a length and/or width of, for example, a metal sheet **404**. By pressing or cutting the apertures **402** into the sheet, the lattice structure **400** may form a series of horizontal or vertical rows of sequential peaks **406** and valleys **408**.

[0036] The apertures defined in the metal sheet **402** may have any shape and dimensions. The perforations may be formed by punching or cutting the third metallic lattice structure **400**. The lattice structure **400** may be described as a plurality of columns of waveform shapes that have troughs (**408**) and peaks (**406**) that alternate with each other side by side. The perforations/apertures **402** of the metallic lattice structure **400** may form any series of shapes or features.

[0037] In an alternative embodiment as seen in FIG. 5, the metallic lattice structure may include an expanded lattice structure **500**. The expanded lattice structure **500** may include, for example, a monolithic lattice structure that is a flattened metal sheet **502** having a repeated series of cuts/punches/holes/openings/apertures **504**, that results in a regular pattern presenting a mesh-like material (e.g., having a rectangular-shaped, square-shaped, diamond-shaped, circular-shaped (see FIG. 7), or other shaped series of patterned apertures). Notably, the apertures **504** may be formed through cutting, stamping, molding or other additive or subtractive manufacturing process. Expanded metal sheets may be relatively stronger than an equivalent weight of, for example, wire mesh, since the material is flattened, allowing the metal to stay in one piece. Further, because expanded metal is never completely cut and reconnected, the material may retain its original strength. In one example, the expanded lattice **500** may include a flattened expanded lattice structure in which the metal sheet is rolled flat or otherwise flattened after being cut to produce an even and flat surface across the openings created by the cutting.

[0038] In an alternative embodiment as seen in FIG. 6, the metallic lattice structure **600** may include an expanded, stretched, corrugated lattice structure **600** in which corrugated openings **602** are formed in the metal sheet **604** between remaining connected segments **606**. In an embodiment, the corrugated lattice structure **600** may be cut and stretched to produce a linear ridged pattern in the metal sheet **602**. The cut and stretched metal sheet **604** results in the formation of a regular pattern (e.g., a diamond-shaped pattern) of mesh-like material. In an embodiment, the metallic

lattice structure may include a wire-mesh lattice structure or a woven mesh lattice structure. The corrugations increase the bending strength of the corrugated lattice structure **600** in the direction perpendicular to the corrugations, but not parallel to the corrugations. In an embodiment, the corrugated lattice structure **600** may be manufactured longer in its strong direction. The wire-mesh lattice structure or woven mesh lattice structure may include a metal mesh that may be, for example, woven, knitted, welded, expanded, sintered, photo-chemically etched, or electroformed, for example.

[0039] In an alternative embodiment, the metallic lattice structure may include a wire netting lattice structure (not specifically shown). The wire netting lattice structure may include, for example, wires that run vertically and are bent into a zig-zag pattern so that each “zig” hooks with a wire immediately on one side and each “zag” with the wire immediately on the other. This may form a diamond pattern.

[0040] FIG. **8** illustrates a flow diagram of an example method **800** of manufacturing a paddle **100**, according to an example of the principles described herein. The method **800** may include, at **802**, applying a (one or more) metallic lattice structure to a portion of the paddle **100**. In an embodiment, the metallic lattice structure may be embodied as any of the first layer **104**, the second layer **106**, and/or the third layer **108**. The method **800** may further include applying an (one or more) intermediate layer to a portion of the paddle **100**. The intermediate layer may be applied as any of the first layer **104**, the second layer **106**, and/or the third layer **108**. The method **800** may further include applying at an (one or more) outer layer to a portion of the paddle **100**. The outer layer may be embodied as, for example, the third layer **108**. Although the method **800** includes application of a number of layers including the metallic lattice structure, the intermediate layer, and/or the outer layer, any quantity or order of layers may be applied to the core **102** of the paddle **100**.

[0041] In an embodiment, the method **800** may include starting with creating composite materials for pickleball paddles **100**. A reinforced metal lattice structure, made up of interconnected elements, may then be embedded within the composite layers to strengthen areas most likely to experience stress during play. Advanced manufacturing techniques such as automated fiber placement or resin infusion may be used to precisely integrate the metal lattice structure within the composite structure. However, traditional composite layup techniques may also be used. This results in paddles **100** that are notably mechanically different from traditional composite structures used in pickleball. The uniqueness of the paddles **100** described herein comes from its ability to improve paddle performance for pickleball. Using the metal lattice structure may increase power, control, maneuverability, and durability, and may enhance the playing experience for a user.

CONCLUSION

[0042] The examples described herein provide a paddle that includes at least one metallic lattice structure. Integrating a reinforced metal lattice structure within composite pickleball paddles or other paddles marks a significant advancement in pickleball equipment design. This approach combines the lightweight nature of composite materials with the strength and durability of metal and results in a high-performing, durable, and satisfying paddle.

[0043] While the present systems and methods are described with respect to the specific examples, it is to be understood that the scope of the present systems and methods are not limited to these specific examples. Since other modifications and changes varied to fit particular operating requirements and environments will be apparent to those skilled in the art, the present systems and methods are not considered limited to the example chosen for purposes of disclosure, and covers all changes and modifications which do not constitute departures from the true spirit and scope of the present systems and methods.

[0044] Although the application describes examples having specific structural features and/or methodological acts, it is to be understood that the claims are not necessarily limited to the specific

features or acts described. Rather, the specific features and acts are merely illustrative of some examples that fall within the scope of the claims of the application.

Claims

1. A paddle comprising: a core; and at least one metallic lattice structure coupled to the core.
2. The paddle of claim 1, wherein the at least one metallic lattice structure is directly coupled to the core or indirectly coupled to the core.
3. The paddle of claim 1, wherein the at least one metallic lattice structure includes a metal or metal alloy.
4. The paddle of claim 1, wherein the at least one metallic lattice structure includes a titanium or titanium alloy.
5. The paddle of claim 1, wherein the at least one metallic lattice structure includes a treated metal.
6. The paddle of claim 1, further comprising at least one intermediate layer coupled to the core and the at least one metallic lattice structure.
7. The paddle of claim 6, further comprising at least one outer layer coupled to the at least one intermediate layer.
8. The paddle of claim 1, further comprising at least one outer layer coupled to the at least one metallic lattice structure.
9. The paddle of claim 1, wherein the at least one metallic lattice structure includes at least one of a wireframe lattice structure, a dimpled lattice structure, a perforated lattice structure, an expanded lattice structure, a flattened expanded lattice structure, a corrugated lattice structure, a wire netting lattice structure, a wire-mesh lattice structure, or a woven mesh lattice structure.
10. The paddle of claim 1, wherein the core includes a honeycomb architecture.
11. A composite layer of a paddle comprising at least one metallic lattice structure.
12. The composite layer of claim 11, wherein the at least one metallic lattice structure is directly coupled to a core or indirectly coupled to the core.
13. The composite layer of claim 11, wherein the at least one metallic lattice structure includes a metal or metal alloy.
14. The composite layer of claim 11, wherein the at least one metallic lattice structure includes a titanium or titanium alloy.
15. The composite layer of claim 11, wherein the at least one metallic lattice structure includes a treated metal.
16. The composite layer of claim 11, further comprising at least one intermediate layer coupled to the at least one metallic lattice structure.
17. The composite layer of claim 16, further comprising at least one outer layer coupled to the at least one intermediate layer.
18. The composite layer of claim 11, further comprising at least one outer layer coupled to the at least one metallic lattice structure.
19. The composite layer of claim 11, wherein the at least one metallic lattice structure includes at least one of a wireframe lattice structure, a dimpled lattice structure, a perforated lattice structure, an expanded lattice structure, a flattened expanded lattice structure, a corrugated lattice structure, a wire netting lattice structure, a wire-mesh lattice structure, or a woven mesh lattice structure.
20. A method of manufacturing a paddle, comprising applying at least one metallic lattice structure to a portion of a paddle.
21. The method of claim 20, wherein applying the at least one metallic lattice structure to a portion of a paddle includes applying the at least one metallic lattice structure directly or indirectly to a core.
22. The method of claim 21, wherein the core includes a honeycomb architecture.
23. The method of claim 20, wherein the at least one metallic lattice structure includes a metal or

metal alloy.

24. The method of claim 20, wherein the at least one metallic lattice structure includes a titanium or titanium alloy.

25. The method of claim 20, wherein the at least one metallic lattice structure includes a treated metal.

26. The method of claim 20, further comprising applying at least one intermediate layer coupled to the at least one metallic lattice structure.

27. The method of claim 26, further comprising applying at least one outer layer to the at least one intermediate layer.

28. The method of claim 20, further comprising applying at least one outer layer to the at least one metallic lattice structure.

29. The method of claim 20, wherein the at least one metallic lattice structure includes at least one of a wireframe lattice structure, a dimpled lattice structure, a perforated lattice structure, an expanded lattice structure, a flattened expanded lattice structure, a corrugated lattice structure, a wire netting lattice structure, a wire-mesh lattice structure, or a woven mesh lattice structure.
